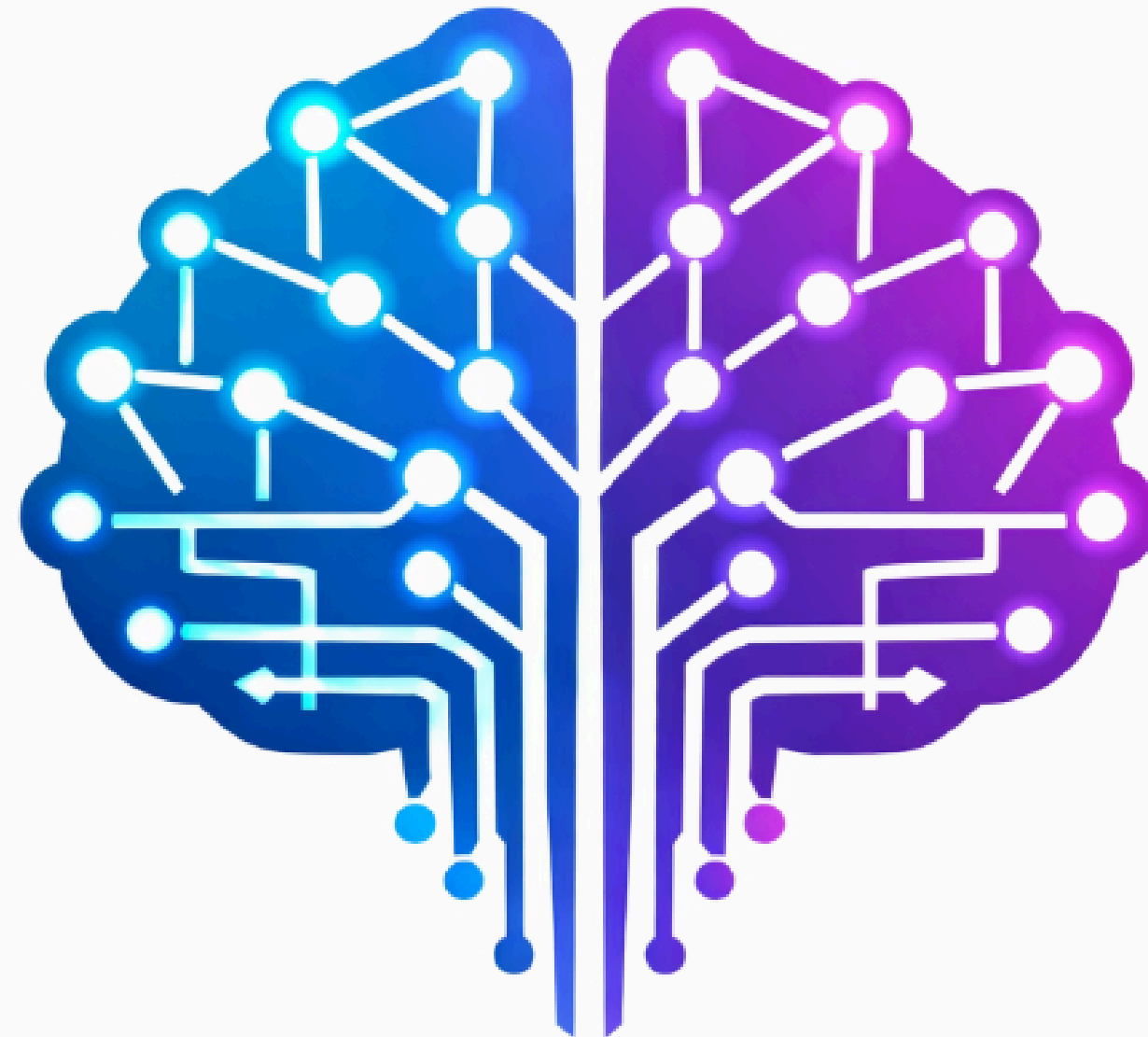


# GENAI FUNDAMENTALS



# **LECTURE:** **LLM LIMITATIONS & FAILURE MODES**

# LLMs Are Powerful but Imperfect

Large Language Models are impressive but not reliable in all situations.

Common limitations include:

- hallucinations
- reasoning mistakes
- outdated knowledge
- sensitivity to prompts

Understanding these limitations is essential for building **safe and reliable AI systems**.

# Hallucinations

LLMs sometimes generate confident but incorrect information.

This phenomenon is called **hallucination**.

Reasons hallucinations occur:

- next-token prediction mechanism
- lack of fact verification
- incomplete training knowledge

# Reasoning Limitations

LLMs can struggle with complex reasoning tasks.

## Examples:

- multi-step logic
- mathematical reasoning
- long chains of deduction

Performance improves with better prompting but is not guaranteed.



# Knowledge Cutoff and Outdated Information

LLMs only know information available during training.

This means they may lack knowledge about:

- recent events
- new research
- updated regulations
- latest technologies

They cannot access **real-time information** unless integrated with external systems.

# Prompt Sensitivity

LLMs can produce very different responses depending on how a question is asked.

Small changes in prompts may cause:

- different reasoning paths
- different answers
- different levels of detail

Prompt design strongly influences model behavior.

# Inconsistent Outputs

LLMs can produce different responses to the same prompt.

Reasons include:

- probabilistic token sampling
- randomness in generation
- temperature settings

This can affect reliability in production systems.



# Key Takeaway

LLMs have powerful capabilities but also clear limitations.

Developers must account for:

- hallucinations
- reasoning errors
- outdated knowledge
- prompt sensitivity
- inconsistent outputs

Reliable GenAI systems require careful design and safeguards.

